

Bracketing Proofs: A Working Guide

How to set up the class, the envelope, and the entropy bound

Companion to *Statistics for Insurance* (Bladt), Ch. 1 & 3 — Theorems 1.18–1.22, 2.8, 3.10, 3.23

Twelve worked examples, a single reusable recipe

This guide has three parts. **Part I** is the method: one recipe, six steps, applied identically every time. **Part II** runs that recipe through fourteen examples — the four from the notes plus ten you have not seen — in a fixed format, so you can read *across* examples and watch the same column do the same job. **Part III** is reference material: master tables, two write-up templates, traps, and an exam plan.

The single most useful sentence in it: *you do not memorise the hypotheses of these theorems — you compute the envelope, demand $\mathbb{E}[F_n^2] = O(1)$, and the hypotheses fall out.*

Contents

Part I — The Method	2
1 What the proof is actually for	2
2 Which theorem	3
3 Step 1 — Building the class	3
4 Step 2 — The envelope	8
5 Step 3 — Computing L^2 norms	9
6 Step 4 — The brackets and the ε^2	10
7 Step 5 — The conditions of Theorem 1.20, fast	12
8 Step 6 — The two archetypes	13
Part II — Twelve Worked Examples	15
Part III — Reference	21
9 Master table	21
10 Write-up template — Theorem 1.20	22
11 Write-up template — Theorem 1.19	22
12 Traps, ranked by how often they cost marks	22
13 Exam plan	23

Part I — The Method

1 What the proof is actually for

You want weak convergence of a process indexed by t , in $\ell^\infty(I)$. That statement is about $\sup_{t \in I}$, i.e. about uncountably many t simultaneously. It splits into two pieces:

$$\text{weak convergence in } \ell^\infty(I) = \underbrace{\text{finite-dimensional convergence}}_{\text{easy: CLT + Cramér–Wold}} + \underbrace{\text{tightness}}_{\text{the hard half — bracketing}}$$

Why the fidis are not enough

Let $U_n \sim \text{Unif}(0, 1)$ and let $\mathbb{X}_n$ be a triangular spike of height 1 and width $1/n$ centred at U_n , zero elsewhere.

- For any fixed t , $P(\mathbb{X}_n(t) \neq 0) \approx 1/n \rightarrow 0$, so $\mathbb{X}_n(t) \rightarrow 0$ in probability. The same holds for any finite collection. **All fidis converge to 0.**
- But $\sup_t \mathbb{X}_n(t) = 1$ for every n . **No convergence in ℓ^∞ .**

The fidis are blind: they inspect finitely many fixed locations and the spike hides between them. Tightness is exactly the requirement that *the process cannot hide a spike between your grid points*.

What bracketing supplies

A bracket $[l_j, u_j]$ is a box in function space. The cell $\mathcal{F}_j$ contains infinitely many f_t , but all of them are trapped, $l_j \leq f \leq u_j$, so

$$\sup_{f, g \in \mathcal{F}_j} |f - g| \leq u_j - l_j \quad \text{pointwise.}$$

Track one bracket instead of infinitely many functions, and pay a penalty of size $\|u_j - l_j\|$. The argument then reads

$$\sup_{t \in I} |\mathbb{G}^{(n)} f_t| \leq \underbrace{\max_{j \leq N} |\mathbb{G}^{(n)}(\text{bracket } j)|}_{\text{finitely many} \Rightarrow \text{union bound}} + \underbrace{\text{oscillation inside cells}}_{\leq \text{bracket width}}.$$

Where $\sqrt{\log N}$ comes from

For finitely many $V_1, \dots, V_N$ with Gaussian-type tails, $P(\max_j V_j > \lambda) \leq N P(V_1 > \lambda) \approx N e^{-\lambda^2/2}$, which is small once $\lambda \approx \sqrt{2 \log N}$. **The price of N competitors is $\sqrt{\log N}$ in the threshold.** That is literally the integrand of the entropy integral. With $N = \infty$ the union bound gives ∞ and the method dies — hence *finitely many brackets*.

Two knobs pull against each other: small ε makes the oscillation term small but $N(\varepsilon)$ large. The entropy integral $\int_0^1 \sqrt{\log N(\varepsilon)} d\varepsilon < \infty$ is precisely the statement that the trade-off resolves in your favour.

2 Which theorem

Theorem	Use when	What to verify
1.18 Glivenko–Cantelli	you only want uniform consistency	$N_{[]}(\varepsilon, \mathcal{F}, L^1(P)) < \infty$ for each ε . Since $\ g\ _{L^1} \leq \ g\ _{L^2}$, an L^2 bound gives this <i>free</i> — always say so.
1.19 Donsker	the class is <i>fixed</i> , rate $\sqrt{n}$, no n in the functions	one condition: $J_{[]}^2(1, \mathcal{F}, L^2(P)) = \int_{(0,1)} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L^2(P))} d\varepsilon < \infty$.
1.20 Changing functions	a bandwidth h_n , threshold u_n , localisation or triangular array appears	(1.6)–(1.9) plus pointwise convergence of the covariances.

Rule of thumb that never fails in these notes

If a bandwidth, a threshold or a k appears anywhere in the estimator, you are in **Theorem 1.20**. A plain i.i.d. empirical process of fixed functions is **Theorem 1.19**.

3 Step 1 — Building the class

The general form: four slots

Every class in this guide — and every class you will be handed — is the same object with four slots filled differently:

The master form

$$\mathcal{F}_n = \left\{ f_{n,t}(\mathbf{x}) = \underbrace{c_n}_{\text{adapter}} \cdot \underbrace{w(\mathbf{x})}_{\text{passenger}} \cdot \underbrace{g_t(\mathbf{x})}_{\text{index carrier}} : t \in \underbrace{I}_{\text{index set}} \right\}$$

- $\mathbf{x}$ is **one observation** — whatever a single data point consists of. That fixes how many arguments $f_{n,t}$ takes.
- c_n is a **number**: the adapter, $\sqrt{n/r_n}$. No t , no data.
- w is the **passenger**: every factor with no t in it. Kernels, censoring indicators, weights, severities, exposures. It rides through untouched.
- g_t is the **index carrier**: the one factor t lives in. It alone decides the archetype.
- I is the **index set**: what you want $\ell^\infty(I)$ convergence on. Its endpoints are decided by the envelope audit, not by taste.

Setting up the class is therefore never invention. It is *reading the summand off the estimator and sorting its factors into the three boxes*. Write the estimator’s summand, then replace each random symbol by a slot letter:

$$\underbrace{\mathbf{1}_{(T_i \leq t)} \delta_i}_{\text{summand, in data}} \longrightarrow \underbrace{\mathbf{1}_{[0,t]}(\tau) d}_{\text{same thing, as a function}} \quad (T_i \rightsquigarrow \tau, \delta_i \rightsquigarrow d).$$

Reading it off the abstract form

The machine only ever speaks one sentence:

$$\mathbb{G}^{(n)} f_{n,t} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(f_{n,t}(X_i) - \mathbb{E} f_{n,t}(X_1) \right).$$

Whatever you were asked to prove can be written as a centred, scaled average of *some* summand $a_t := w g_t$, divided by *some* r_n :

$$\mathbb{X}_n(t) = \sqrt{r_n} \left(\frac{1}{r_n} \sum_{i=1}^n a_t(X_i) - \mathbb{E}[\cdot] \right).$$

Matching the two is one line of algebra — there is no modelling decision in it:

The identification

$$\sqrt{r_n} \left(\frac{1}{r_n} \sum_i a_t(X_i) - \mathbb{E} \right) = \frac{1}{\sqrt{r_n}} \sum_i (a_t(X_i) - \mathbb{E} a_t(X_1)) = \frac{1}{\sqrt{n}} \sum_i \underbrace{\left(\sqrt{\frac{n}{r_n}} a_t(X_i) - \mathbb{E}[\cdot] \right)}_{= f_{n,t}} = \mathbb{G}^{(n)} f_{n,t}$$

$$f_{n,t} = \sqrt{n/r_n} \cdot a_t = c_n w g_t$$

So there are exactly two things to extract from the problem: **the summand** a_t (sort its factors into w and g_t) and **the divisor** r_n . The $\sqrt{n/r_n}$ is then forced.

The identity runs both ways, which gives a free check: *a class determines the process it represents*, via

$$r_n = \frac{n}{c_n^2}, \quad \text{so} \quad \mathbb{G}^{(n)} f_{n,t} = \sqrt{r_n} \left(\frac{1}{r_n} \sum_i a_t(X_i) - \mathbb{E} \right).$$

E.g. $c_n = h^{-1/2} \Rightarrow r_n = nh$; $c_n = \bar{F}(u_n)^{-1/2} \Rightarrow r_n = n\bar{F}(u_n) = k$; $c_n = 1 \Rightarrow r_n = n$. If the r_n you recover is not the rate in the statement you were asked to prove, your c_n is wrong — catch it here, before any bracketing.

Reading the slots off: three examples

Write the estimator's summand. Then ask three questions in order: *what divides the sum?* $\rightarrow c_n$. *Which factors have no t ?* $\rightarrow w$. *Where does t live?* $\rightarrow g_t$.

(i) **Censored claims.** $\frac{1}{n} \sum_i \mathbf{1}_{(T_i \leq t)} \delta_i$, one observation is the pair (T_i, δ_i) .

divisor is n	every observation counts, $r_n = n$	$c_n = \sqrt{n/n} = 1$
δ_i has no t	a passenger	$w(\tau, d) = d$
t in the indicator	non-decreasing, archetype A	$g_t(\tau) = \mathbf{1}_{[0,t]}(\tau)$

$$\mathcal{F} = \{f_t(\tau, d) = \mathbf{1}_{[0,t]}(\tau) d\}, \quad I = [0, \infty).$$

Lesson: an extra factor in the estimator is not an obstacle — it is a passenger, and since $d^2 = d$ it costs nothing in the L^2 step.

(ii) **Tail empirical process.** $\frac{1}{k} \sum_i \mathbf{1}_{(Z_i > u_n t)}$, threshold u_n , $k = n\bar{F}(u_n)$.

divisor is k	only exceedances count, $r_n = c_n = \sqrt{n/k} = \bar{F}(u_n)^{-1/2}$ $k = n\bar{F}(u_n)$	
nothing else	no passenger	$w \equiv 1$
t in the indicator	non-increasing, archetype A	$g_t(z) = \mathbf{1}_{(z > u_n t)}$
$\mathcal{F}_n = \{f_{n,t}(z) = \bar{F}(u_n)^{-1/2} \mathbf{1}_{(z > u_n t)}\}, \quad I = [a, b], a > 0.$		

Lesson: the divisor of the estimator is r_n . And u_n sits inside g_t as part of the cut point — it is not an index. The restriction $a > 0$ is not read off the problem; it is forced later by the envelope audit.

(iii) **Excess-of-loss layer at a fixed policy age.** $\frac{1}{nh} \sum_i (C_i - t)_+ K(h^{-1}(a - A_i))$, one observation is (A_i, C_i) .

divisor is nh	only the window counts, $r_n = nh$	$c_n = \sqrt{n/nh} = h^{-1/2}$
K has no t	a passenger	$w(c, \alpha) = K(h^{-1}(a - \alpha))$
t in $(C_i - t)_+$	not an indicator — 1-Lipschitz, archetype B	$g_t(c) = (c - t)_+$
$\mathcal{F}_n = \{f_{n,t}(c, \alpha) = h^{-1/2}(c - t)_+ K(h^{-1}(a - \alpha))\}, \quad I = [0, \tau].$		

Lesson: the carrier need not be an indicator. Look at what g_t is — that alone decides whether the grid will be ε^2 (A) or ε (B), independently of which theorem you are in.

The same three questions handle everything else: a conditional survival function puts $\mathbf{1}_{(y > t)}$ in the carrier; an IPCW weight $1/\bar{G}(\tau)$ or a severity C_i goes in the passenger; two localisations multiply their c_n 's; a bivariate index makes $g_{s,t}$ a product of indicators.

Sanity checks

1. **Does the class reproduce the process?** Verify $\frac{c_n}{\sqrt{n}} = \frac{1}{\sqrt{r_n}}$, so $\mathbb{G}^{(n)} f_{n,t} = \sqrt{r_n}(\hat{\theta}_n(t) - \mathbb{E})$.
2. **Is t confined to g_t ?** No t in c_n , none in w .
3. **Does $f_{n,t}$ take the right number of arguments?** One slot per component of a single observation.
4. **Is $\mathbb{E}[F_n^2]$ a finite positive constant?** If not, fix c_n or shrink I .

The adapter c_n : what it is for

The machine is hard-wired to divide by $\sqrt{n}$:

$$\mathbb{G}^{(n)} f_{n,t} = \frac{1}{\sqrt{n}} \sum_{i=1}^n (f_{n,t}(X_i) - \mathbb{E} f_{n,t}(X_1)),$$

but a localised estimator does not use all n observations. A kernel estimator at x ignores every X_i outside the window; a tail estimator listens only to the $\approx k$ observations above u_n . The honest sample size is r_n , and the right scaling is $\sqrt{r_n}$ — which is why the theorems state rates $\sqrt{nh}$, $\sqrt{k}$, $\sqrt{k_n h_n}$. The mismatch is absorbed into the function:

$$\boxed{c_n = \sqrt{n/r_n}} \iff c_n \text{ is whatever makes } \text{Var}(f_{n,t}(X_1)) = O(1),$$

$$\text{because } \frac{1}{\sqrt{n}} \cdot c_n = \frac{1}{\sqrt{n}} \cdot \frac{\sqrt{n}}{\sqrt{r_n}} = \frac{1}{\sqrt{r_n}}.$$

c_n is not statistics, it is an adapter. The two descriptions agree because $g_t w$ is supported on an event of probability $\asymp r_n/n$, so $c_n^2 \mathbb{E}[(w g_t)^2] \asymp 1$. The symptom that you chose it correctly is condition (1.6).

Finding r_n

$r_n = n \times P(\text{an observation contributes}).$

Localisation	Contributes when	$P(\cdot)$	r_n
none	always	1	n
kernel window at x , bandwidth h	$ X_i - x \lesssim h$	$\approx h$	nh
threshold u_n	$Z_i > u_n$	$\bar{F}(u_n)$	$k = n\bar{F}(u_n)$
both	both	$h \bar{F}(u_n^x)$	$k_n h_n$

The fourteen examples, as filled-in slots

Example	c_n	passenger w	carrier g_t	I
1 empirical df	1	1	$\mathbf{1}_{[0,t]}(z)$	$[0, \infty)$
2 bivariate df	1	1	$\mathbf{1}_{[0,s]}(z) \mathbf{1}_{[0,t]}(w)$	$[0, \infty)^2$
3 censored H_1	1	d	$\mathbf{1}_{[0,t]}(\tau)$	$[0, \infty)$
4 IPCW	1	$d/\bar{G}(\tau)$	$\mathbf{1}_{[0,t]}(\tau)$	$[0, \tau^*], \bar{G}(\tau^*) > 0$
5 kernel cdf	$h^{-1/2}$	$K(h^{-1}(x - z))$	$\mathbf{1}_{[0,t]}(y)$	$[0, \infty)$
6 kernel survival	$h^{-1/2}$	$K(h^{-1}(x - z))$	$\mathbf{1}_{(y>t)}$	$[0, \tau]$
7 tail process	$\bar{F}(u_n)^{-1/2}$	1	$\mathbf{1}_{(z>u_n t)}$	$[a, b], a > 0$
8 cond. tail	$(h_n \bar{F}(u_n^x))^{-1/2}$	$K(h_n^{-1}(z - x))$	$\mathbf{1}_{(y>u_n^x t)}$	$[a, b], a > 0$
9 XL layer	$h^{-1/2}$	$K(h^{-1}(a - \alpha))$	$(c - t)_+$	$[0, \tau]$
10 tail moment	$\bar{F}(u_n)^{-1/2}$	z/u_n	$\mathbf{1}_{(z>u_n t)}$	$[a, b], a > 0$
11 exp. tilts	1	1	$e^{-\theta z}$	$[0, 1]$
12 affine	1	1	$a + bz$	$[-1, 1]^2$
13 stop-loss	1	c	$\mathbf{1}_{(c>t)}$	$[0, \infty)$
14 cond. stop-loss	$h^{-1/2}$	$c K(h^{-1}(a - \alpha))$	$\mathbf{1}_{(c>t)}$	$[0, \infty)$

Read down the columns. The c_n column is determined entirely by the localisation; the w column collects whatever else the estimator happened to carry; the g_t column is the only one that affects the bracketing; and the I column is decided by the envelope audit.

The envelope and the controlling function, in general

Because c_n and w carry no t , the sup passes straight through them:

$$F_n = \sup_{t \in I} |f_{n,t}| = c_n |w| \cdot \sup_{t \in I} |g_t| = c_n |w| g_{t^*},$$

where t^* is the endpoint of I at which the monotone family g_t is largest. That is the whole of Step 2.

Now the payoff. Define the **controlling function** as the squared norm of a single member of the class, viewed as a function of t :

You never have to guess the controlling function

$$\Lambda_n(t) := \|f_{n,t}\|_{L^2(P)}^2 = c_n^2 \mathbb{E}[w(X_1)^2 g_t(X_1)^2]$$

If the carrier g_t is an **indicator** of a family of sets nested in t (archetype A), then $g_t^2 = g_t$, the sets are nested, and for a cell $[t_{j-1}, t_j]$

$$\Delta_j = \|u_j - l_j\|_{L^2(P)}^2 = |\Lambda_n(t_j) - \Lambda_n(t_{j-1})|, \quad \Delta_{\text{env}} = \|F_n\|_{L^2(P)}^2 = \Lambda_n(t^*).$$

Hence the grid condition and the count are, *always*,

$$\Delta_j \leq \varepsilon^2 \Lambda_n(t^*) \quad \implies \quad m \leq \left\lceil \frac{2 \Lambda_n(t^*)}{\varepsilon^2 \Lambda_n(t^*)} \right\rceil = \left\lceil \frac{2}{\varepsilon^2} \right\rceil.$$

$\Lambda_n(t^*)$ cancels. **Compute Λ_n once and every remaining step is filled in.**

Λ_n is automatically monotone (nested sets), automatically non-negative, and its total variation on I is at most $\Lambda_n(t^*)$ — which is exactly the “càdlàg, non-decreasing and bounded” clause the notes invoke when asserting the grid exists. And every “controlling function” named anywhere in this guide is just $\lim_n \Lambda_n$:

Example	$\Lambda_n(t) = c_n^2 \mathbb{E}[w^2 g_t^2]$	$\lim_n \Lambda_n(t)$
1 empirical df	$\mathbb{E}[\mathbf{1}_{(Z_1 \leq t)}]$	$F(t)$
3 censored	$\mathbb{E}[\mathbf{1}_{(T_1 \leq t)} \delta_1]$	$H_1(t)$
4 IPCW	$\mathbb{E}[\mathbf{1}_{(T_1 \leq t)} \delta_1 / \bar{G}(T_1)^2]$	a weighted sub-distribution function
5 kernel cdf	$\mathbb{E}[h^{-1} K^2 \mathbf{1}_{(Y_1 < t)}]$	$F(t x) f(x) \int K^2$
6 kernel survival	$\mathbb{E}[h^{-1} K^2 \mathbf{1}_{(Y_1 > t)}]$	$S(t x) f(x) \int K^2$
7 tail process	$\bar{F}(u_n t) / \bar{F}(u_n)$	$T(t) = t^{-1/\gamma}$
8 cond. tail	$(h_n \bar{F})^{-1} \mathbb{E}[K^2 \mathbf{1}_{(Y_1 > u_n^* t)}]$	$T^x(t) g(x) \int K^2$
10 tail moment	$\bar{F}(u_n)^{-1} \mathbb{E}[(Z_1 / u_n)^2 \mathbf{1}_{(Z_1 > u_n t)}]$	$M(t) = \frac{\beta}{\beta-2} t^{2-\beta}$
13 stop-loss	$\mathbb{E}[C_1^2 \mathbf{1}_{(C_1 > t)}]$	itself (a restricted 2nd moment)
14 cond. stop-loss	$\mathbb{E}[h^{-1} K^2 C_1^2 \mathbf{1}_{(C_1 > t)}]$	$L_2(t a) f_A(a) \int K^2$

This is why the controlling function stops needing to be a probability: the passenger w squares into the measure that weighs the gap. **The only properties ever used are that Λ_n is monotone and finite on I .**

The archetype-B analogue

If the carrier is not an indicator, $g_t^2 \neq g_t$, increments of Λ_n are the wrong object, and you use the Lipschitz majorant instead. Suppose $|g_t(\mathbf{x}) - g_s(\mathbf{x})| \leq |t - s| \mu(\mathbf{x})$. Then

$$\|u_j - l_j\|_{L^2(P)} \leq (t_j - t_{j-1}) c_n \|w\mu\|_{L^2(P)}, \quad \frac{\|u_j - l_j\|}{\|F_n\|} \leq \frac{(t_j - t_{j-1}) \|w\mu\|_{L^2(P)}}{\|w g_{t^*}\|_{L^2(P)}},$$

so c_n still cancels, the mesh condition is $t_j - t_{j-1} \leq \varepsilon \|w g_{t^*}\|_{L^2(P)} / \|w\mu\|_{L^2(P)}$, and with $|I|$ the length of the index set, $m \leq \lceil C/\varepsilon \rceil$. Example 9 is this with $\mu \equiv 1$, $w = K$, giving $C = |I|/\sqrt{\mathbb{E}[C^2 \mid A = a]}$; Example 11 is this with $w \equiv 1$, $\mu(z) = z$.

Two habits that save marks

(i) Strip the t -free parts first. A Nadaraya–Watson estimator is a ratio; its denominator is not a process in t . Handle it by joint fidi convergence and Slutsky (Theorem 1.8), and bracket only the numerator. Theorem 2.8 does exactly this.

(ii) Index by t only. h, u_n, x, K are frozen *inside* the function — they belong to c_n and w , never to the index. And never let t appear inside c_n .

4 Step 2 — The envelope

$$F_n(\cdot) = \sup_{t \in I} |f_{n,t}(\cdot)|$$

A *function*, not a number: the arguments are held fixed while the sup runs over t . An envelope need only *dominate* the class, so you may replace it by anything larger and simpler.

Computing it: exactly two moves

1. **Factors with no t come outside the sup, unchanged.** $(c_n, K(h^{-1}(x - z)), \delta, 1/\overline{G}(\tau), z/u_n, \dots)$
2. **For the t -dependent factor, decide the direction of monotonicity and evaluate at the endpoint of I that makes it largest.**

t -dependent factor	direction in t	sup attained at
$\mathbf{1}_{[0,t]}(y)$	non-decreasing	right endpoint of I (often = 1)
$\mathbf{1}_{(y>t)}, \mathbf{1}_{(y>u_n t)}$	non-increasing	left endpoint of I
$(c - t)_+$	non-increasing	left endpoint; value $(c - t_{\min})_+$
$e^{-\theta z}, z \geq 0$	non-increasing in θ	left endpoint

The envelope is where the theorem's hypotheses come from

Condition (1.6), $\mathbb{E}[F_n^2] = O(1)$, is an *audit*. It fails in exactly three ways, and each failure tells you what to change:

- $\rightarrow 0$: c_n too small. The limit process would be degenerate at 0.
- $\rightarrow \infty$ **because of** c_n : c_n too large.
- $\rightarrow \infty$ **because of the envelope's shape**: the index set I reaches a boundary where mass vanishes or a weight blows up. *Shrink* I , or impose the moment condition the computation is demanding.

What the envelope forced, in each example

Example	Envelope	Hypothesis it forces
empirical df	1	nothing
censored H_1	1	nothing
kernel conditional df	$h^{-1/2}K$	$f(x) > 0$, f continuous at x
IPCW weighting	$1/\bar{G}(\tau)$	$\bar{G}(\tau^*) > 0$: stop before risk set empties
tail empirical process	$\bar{F}(u_n)^{-1/2}\mathbf{1}_{(z > u_n a)}$	I bounded away from 0
excess-of-loss layer	$h^{-1/2}cK$	$\mathbb{E}[C^2 \mid A = a] < \infty$
tail moment	$\bar{F}(u_n)^{-1/2}(z/u_n)\mathbf{1}_{(z > u_n a)}$	$\gamma < 1/2$ (finite tail variance)
stop-loss	c	$\mathbb{E}[C_1^2] < \infty$ — no index set can fix this
conditional stop-loss	$h^{-1/2}cK$	$\mathbb{E}[C_1^2 \mid A_1 = a] < \infty$

5 Step 3 — Computing L^2 norms

Throughout, the norm is taken under the **law of the data**, not Lebesgue measure:

$$\|g\|_{L^2(P)} = \left(\int g^2 dP \right)^{1/2} = (\mathbb{E}[g(X_1)^2])^{1/2}, \quad \text{so} \quad \|g\|_{L^2(P)}^2 = \mathbb{E}[g(X_1)^2].$$

That identity is a *definition*, not a theorem — and the square root it carries is the origin of every ε^2 in this subject.

Five rules; every computation in Part II is one of them

R1 (scaling). $\|cg\| = |c| \|g\|$. Consequence: c_n always cancels in the ratio $\|u_j - l_j\| / \|F_n\|$. Never carry $h^{-1/2}$ through the algebra.

R2 (indicators — the square-root rule). An indicator squared is itself, so for $A \subseteq B$,

$$\|\mathbf{1}_B - \mathbf{1}_A\|_{L^2(P)}^2 = P(B \setminus A), \quad \|\mathbf{1}_B - \mathbf{1}_A\|_{L^2(P)} = \sqrt{P(B \setminus A)}.$$

Concretely $\|\mathbf{1}_{[0,t]} - \mathbf{1}_{[0,s]}\|^2 = F(t-) - F(s-)$ for $s < t$. **This is the whole source of ε^2 :** to make a distance ε you need a mass ε^2 .

R3 (products — condition on what you can). If $f_{n,t}(y, z) = \mathbf{1}_{A_t}(y)w(z)$ then $\mathbb{E}[f_{n,t}^2] = \mathbb{E}[w(X_1)^2 P(Y_1 \in A_t \mid X_1)]$. With a kernel weight this converges to $P(Y_1 \in A_t \mid X_1 = x) \cdot f(x) \int K^2$: (mass in the response) $\times$ (a constant from the localisation). The factorisation is why the localisation constants cancel.

R4 (Lipschitz in the parameter). If $|f_\theta(z) - f_{\theta'}(z)| \leq |\theta - \theta'| m(z)$ with $m \in L^2(P)$, then $\|f_\theta - f_{\theta'}\| \leq |\theta - \theta'| \|m\|$. Bracket size is *linear* in the mesh — no square root.

R5 (corners). For families monotone or linear in the index, inf and sup over a cell sit at the cell's corners, so l_j, u_j are the corner functions and $u_j - l_j$ is explicit. For $a + bz$ on a rectangle: $u - l \leq \Delta_a + \Delta_b |z|$.

Two standard limits

Kernel window (f continuous at $x, z = x + hu$) **Regular variation** ($\bar{F} \in RV_{-1/\gamma}$, uniformly on compacts)

$$\begin{aligned} \mathbb{E} \left[h^{-1} K^p \left(\frac{x - X_1}{h} \right) \right] &= \int K^p(u) f(x + hu) du \\ &= f(x) \int K^p + O(h) \end{aligned} \quad \begin{aligned} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} &\longrightarrow t^{-1/\gamma} = T(t) \\ \text{(uniformity is what makes the } o(1) \text{ independent of the partition,} \\ \text{which is what lets you skip (1.8))} \end{aligned}$$

6 Step 4 — The brackets and the ε^2

The bracketing number Theorem 1.20 actually uses

Not Definition 1.16. Theorem 1.20 asks for a *partition* $\mathcal{F}_n = \bigcup_{j=1}^{N_\varepsilon} \mathcal{F}_{\varepsilon j}^n$ with

$$\mathbb{E} \left[\sup_{f_{n,t}, f_{n,s} \in \mathcal{F}_{\varepsilon j}^n} |f_{n,t}(X_1) - f_{n,s}(X_1)|^2 \right] \leq \varepsilon^2 \|F_n(X_1)\|_{L^2(P)}^2.$$

Two features, both load-bearing:

1. **Brackets produce such a partition.** If $l_j \leq f \leq u_j$ on $\mathcal{F}_j^n$ then $\sup_{f,g} |f - g| \leq u_j - l_j$ pointwise, so a bracket bound implies the oscillation bound. *Write this sentence once and then work with brackets.*
2. **The tolerance is $\varepsilon \|F_n\|$, not ε .** A *relative* accuracy. This is what makes the scheme invariant to c_n : multiply $f_{n,t}$ by anything and both sides scale identically. Without it the bracket count would grow with n (e.g. like h^{-1}) and the entropy integral would never go to zero.

The derivation, in discovery order

A. Write a generic bracket; do not choose the grid yet

Cut I at unspecified $t_0 < \dots < t_m$. On cell j take the two *endpoint* functions.

- non-decreasing family: $l_j = c_n \mathbf{1}_{[0, t_{j-1})} w$, $u_j = c_n \mathbf{1}_{[0, t_j)} w$, so $u_j - l_j = c_n \mathbf{1}_{[t_{j-1}, t_j)} w$;
- non-increasing family: roles swap, $u_j - l_j = c_n \mathbf{1}_{[t_{j-1}, t_j)} w$ again.

Either way the difference is an indicator of the gap times the t -free factor $c_n w$.

B. Compute the norm as a function of the unknown gap

Write Δ_j for the (possibly weighted) mass of the gap and Δ_{env} for the mass of the envelope's event. By R2–R3 *the same constant appears in both*:

$$\|u_j - l_j\|^2 = \Delta_j \cdot (\text{const}), \quad \|F_n\|^2 = \Delta_{\text{env}} \cdot (\text{same const}),$$

hence

$$\frac{\|u_j - l_j\|}{\|F_n\|} = \sqrt{\frac{\Delta_j}{\Delta_{\text{env}}}} (1 + o(1)).$$

C. Invert — the step the write-ups hide

$$\|u_j - l_j\| \leq \varepsilon \|F_n\| \iff \boxed{\Delta_j \leq \varepsilon^2 \Delta_{\text{env}}}$$

There is the ε^2 : it is the *square of the target*, because the L^2 norm of an indicator difference is the square *root* of a mass. You compute a distance, then read off a mass. The notes present the mass first because that is the order a *finished* proof reads in.

D. Count

The cells partition a total budget of at most Δ_{env} , each eating $\varepsilon^2 \Delta_{\text{env}}$, so you need about ε^{-2} of them. Δ_{env} **cancels a second time**, which is why every proof in the notes — for the df, the conditional df, the tail, the conditional tail — lands on the same bound:

$$N_{[]}(\varepsilon \|F_n\|_{L^2(P)}, \mathcal{F}_n, L^2(P)) \leq m \leq \left\lceil \frac{C}{\varepsilon^2} \right\rceil.$$

The budget recipe (recommended)

Rather than carrying ε from the start, do the algebra with a generic per-cell budget δ and substitute at the end.

Three lines that cover every example in this guide

1. Compute $\varphi(\delta) := \frac{\|u_j - l_j\|_{L^2(P)}}{\|F_n\|_{L^2(P)}}$ when each cell has budget δ . For archetype A this is $\sqrt{\delta/\Lambda_n(t^*)}$ with $\Lambda_n(t) = \|f_{n,t}\|_{L^2(P)}^2$ — see §3.
2. Solve $\varphi(\delta) = \varepsilon$ for δ . That is your grid condition.
3. Count $m \approx \text{Total}/\delta(\varepsilon)$, where Total is the total budget on I .

Example	budget is	$\varphi(\delta)$	solve	m
empirical df	mass	$\sqrt{\delta}$	$\delta = \varepsilon^2$	$1/\varepsilon^2$
height-3 toy	mass	$3\sqrt{\delta}$	$\delta = \varepsilon^2/9$	$9/\varepsilon^2$
censored H_1	mass	$\sqrt{\delta}$	$\delta = \varepsilon^2$	$1/\varepsilon^2$
IPCW	mass	$\sqrt{\delta/\overline{G}(\tau^*)}$	$\delta = \varepsilon^2 \overline{G}(\tau^*)^2$	$\overline{G}(\tau^*)^{-2}/\varepsilon^2$
tail process	mass under T	$\sqrt{\delta/T(a)}$	$\delta = \varepsilon^2 T(a)$	$1/\varepsilon^2$
tail moment	mass under M	$\sqrt{\delta/M(a)}$	$\delta = \varepsilon^2 M(a)$	$1/\varepsilon^2$
excess-of-loss	mesh	$\delta/\sqrt{\mathbb{E}[C^2 a]}$	$\delta = \varepsilon\sqrt{\mathbb{E}[C^2 a]}$	$\tau/(\varepsilon\sqrt{\cdot})$
exponential tilts	mesh	$\delta\ Z\ _{L^2}$	$\delta = \varepsilon/\ Z\ _{L^2}$	$\ Z\ _{L^2}/\varepsilon$

Everything you could be confused about now sits in one column, φ : *square root* $\Rightarrow \delta \propto \varepsilon^2$; *linear* $\Rightarrow \delta \propto \varepsilon$. The constants inside φ are whatever failed to cancel against $\|F_n\|$.

Two details worth having a sentence ready for

Where the factor 2 comes from. Build the grid greedily: $t_0 = 0$ and $t_j = \sup\{t > t_{j-1} : F(t-) - F(t_{j-1}) \leq \delta\}$. Then gap j carries mass $\leq \delta$, and by maximality $F(t_j) - F(t_{j-1}) > \delta$ — the gap *together with the atom at t_j* exceeds δ . Summing over every second index, $m \leq 2/\delta$. With F continuous there are no atoms and $m \leq 1/\delta + 1$.

The Total must be finite. This is the entire content of “possible because F is càdlàg, non-decreasing and bounded by 1”. It is exactly what fails for the tail process when $a \downarrow 0$: $T(0) = \infty$, no finite grid exists. Step 3 of the recipe is where a badly chosen index set announces itself, just as (1.6) is where a badly chosen c_n does.

The last line: integrating

With $N \leq C\varepsilon^{-p}$, using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and $\varepsilon = e^{-v}$,

$$\int_0^1 \sqrt{\log N(\varepsilon)} d\varepsilon \leq \sqrt{\log C} + \sqrt{p} \int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon = \sqrt{\log C} + \sqrt{p} \Gamma(3/2) < \infty, \quad \Gamma(3/2) = \frac{\sqrt{\pi}}{2}.$$

- For **Theorem 1.19** that finiteness is the hypothesis. Done.
- For **Theorem 1.20**, condition (1.9) wants $\int_{[0,\delta_n]} \rightarrow 0$ for every $\delta_n \rightarrow 0$. This follows from finiteness for free, by *absolute continuity of the Lebesgue integral*. Say that sentence; do not re-derive it.

How much room you have

The integral converges whenever $\log N_{[\cdot]}(\varepsilon) = O(\varepsilon^{-2+\eta})$ for some $\eta > 0$. A polynomial bound gives $\log N \asymp \log(1/\varepsilon)$ — nowhere near the boundary. **Any polynomial bound at all is enough; never optimise p , never chase a constant.** Write “for some constant $C < \infty$ ” and move on.

7 Step 5 — The conditions of Theorem 1.20, fast

(1.6) $\mathbb{E}[F_n^2] = O(1)$. The normalisation audit; see §4. One line with the kernel or regular-variation lemma.

(1.7) **Lindeberg**. One trick, every time: on $\{F_n > \eta\sqrt{n}\}$ we have $1 \leq F_n/(\eta\sqrt{n})$, so

$$\mathbb{E}[F_n^2 \mathbf{1}_{(F_n > \eta\sqrt{n})}] \leq \frac{\mathbb{E}[F_n^3]}{\eta\sqrt{n}} = O\left(\frac{1}{\eta\sqrt{r_n}}\right) \rightarrow 0 \iff r_n \rightarrow \infty.$$

The scaling holds because $F_n = c_n G_n$ with $\mathbb{E}[G_n^3]/\mathbb{E}[G_n^2] = O(1)$, so $\mathbb{E}[F_n^3]/\mathbb{E}[F_n^2] = c_n \cdot O(1) = \sqrt{n}/r_n \cdot O(1)$.

Theorem	$\mathbb{E}[F_n^3]$	bound	vanishes iff
2.8	$h^{-1/2}(f(x) \int K^3 + O(h))$	$\frac{f(x) \int K^3}{\eta\sqrt{nh}}$	$nh \rightarrow \infty$
3.10	$\bar{F}(u_n)^{-3/2} \bar{F}(u_n a)$	$\frac{a^{-1/\gamma}}{\eta\sqrt{k}}$	$k \rightarrow \infty$
3.23	$(h_n \bar{F}(u_n^x))^{-3/2}(\dots)$	$O(1/(\eta\sqrt{k_n h_n}))$	$k_n h_n \rightarrow \infty$

When the third-moment shortcut over-assumes

It needs a finite third moment of the envelope. For heavy tails that can demand strictly more than the problem gives (e.g. $\gamma < 1/3$ instead of $\gamma < 1/2$). Then *drop it* and compute directly: the truncation level always diverges. See Example 10, where $\mathbb{E}[F_n^2 \mathbf{1}_{(F_n > \eta\sqrt{n})}] \approx M(\eta\sqrt{k}) \rightarrow 0$ under $\gamma < 1/2$ alone.

(1.8) **Oscillation**. Usually free. The last sentence of Theorem 1.20: *if the partition points in t can be chosen independently of n , (1.8) is unnecessary*. They can, because the grid is built from a **limit** object — $F, F(\cdot | x), T, T^x, H_1, M$ — which does not move with n .

Sentence to memorise verbatim

“The $o(1)$ term above is uniform in the partition by [uniform convergence / regular variation / Thm 3.20], so the partition points may be chosen independently of n ; by the final statement of Theorem 1.20, condition (1.8) need not be verified.”

Be precise about the distinction: the *bracket functions* l_j, u_j certainly depend on n (they contain u_n, h). What must be n -free is the *partition points* t_j in the index set. That is the sentence’s content and the thing examiners probe.

(1.9) **Entropy integral**. §4.

Covariance. Theorem 1.20 delivers a tight Gaussian limit *provided* $\mathbb{E}[f_{n,t} f_{n,s}] - \mathbb{E}f_{n,t} \mathbb{E}f_{n,s}$ converges pointwise. This is not bracketing work — it is the finite-dimensional step, and it is where the actual limit comes from. Keep it visibly separate in your write-up.

8 Step 6 — The two archetypes

	A. Monotone / indicator	B. Smooth in a parameter
looks like	$\mathbf{1}_{[0,t]}(y), \mathbf{1}_{(y>u_n t)}, \mathbf{1}_{(y\leq t)}\delta$	$\theta z, a + bz, e^{-\theta z}, (c - t)_+, y \wedge t$
bracket	two neighbouring indicators	min / max over the cell's corners
$u_j - l_j$ is	an indicator of the gap	$\leq \text{mesh} \times m(z)$
L^2 size	$\sqrt{\text{mass of gap}}$ — square root	$\text{mesh} \times \ m\ _{L^2(P)}$ — linear
grid	$\text{mass} \leq \varepsilon^2$	$\text{mesh} \leq \varepsilon / \ m\ _{L^2(P)}$
count	$N \leq \lceil C/\varepsilon^2 \rceil$	$N \leq (C/\varepsilon)^d, d = \dim \theta$
needs	controlling function monotone, bounded	an L^2 Lipschitz majorant m

Three confusions to pre-empt

(i) **Theorem 1.20** $\neq \varepsilon^2$. Example 9 is a changing class (bandwidth, $\sqrt{nh}$ rate) with an ε -grid. Example 1 is a fixed class with an ε^2 -grid. *What decides the exponent is what $u_j - l_j$ turns out to be, not which theorem you are in.*

(ii) **Two different sources of ε^{-2}** . Archetype A with $d = 1$ gives ε^{-2} from the square root. Example 12b (affine class) gives ε^{-2} from having *two parameters*. If you can say which mechanism produced the exponent in a given problem, you understand this section.

(iii) **Monotone $\neq$ archetype A**. $e^{-\theta z}$ is monotone in θ , so you get the sandwich for free — but $u - l$ is not an indicator, so the *size* is governed by R4, not R2. Always look at what $u_j - l_j$ is.

The deepest single fact in the method

What makes a class archetype A is **where t sits**, not what the passenger is. If t lives inside an indicator, $u_j - l_j$ is supported on a gap and you get a square root. The passenger merely changes *which measure* the gap is weighed in — and the controlling function stops needing to be a probability. It only has to be *monotone and finite on I* .

Part II — Fourteen Worked Examples

Each example uses the same headings, so you can read across. Examples 1, 5, 7, 8 are the four in the notes; the rest are new scenarios of the kind an exam sets. Examples 13 and 14 are a matched pair — the same estimator with and without localisation — and are the fastest way to see what the $\|F_n\|$ in Theorem 1.20's tolerance is actually for.

Example 1 — Empirical distribution function Theorem 1.22; archetype A; the base case

Process. $\sqrt{n}(F^{(n)} - F)$ in $\ell^\infty([0, \infty))$, $Z_i \geq 0$ i.i.d.

Rate & class. Every observation counts: $r_n = n$, $c_n = 1$. Fixed class $\Rightarrow$ **Theorem 1.19**.

$$\mathcal{F} = \{f_t(z) = \mathbf{1}_{[0,t]}(z) : t \in [0, \infty)\}$$

Envelope. $t \mapsto \mathbf{1}_{[0,t]}(z)$ non-decreasing, sup over t equals 1 for every z : $F \equiv 1$, $\mathbb{E}[F^2] = 1$.

Brackets. $l_j = \mathbf{1}_{[0,t_{j-1}]}$, $u_j = \mathbf{1}_{[0,t_j]}$, so $u_j - l_j = \mathbf{1}_{[t_{j-1}, t_j]}$.

L^2 . By R2, $\|u_j - l_j\|^2 = F(t_j) - F(t_{j-1}) =: \Delta_j$; $\varphi(\delta) = \sqrt{\delta}$.

Grid & count. $\Delta_j < \varepsilon^2$ is possible because F is càdlàg, non-decreasing and bounded by 1; Total = 1, so $m \leq \lceil 2/\varepsilon^2 \rceil$ and $N_{[]}(\varepsilon, \mathcal{F}, L^2(P)) \leq \lceil 2/\varepsilon^2 \rceil$.

Finish. $J_{[]}(\mathbf{1}, \mathcal{F}, L^2(P)) \leq \int_{(0,1]} \sqrt{\log(2/\varepsilon^2 + 1)} d\varepsilon < \infty$; P -Donsker, limit $B \circ F$. Free line: $L^1 \leq L^2$ gives Glivenko–Cantelli (Thm 1.18).

Example 2 — Bivariate empirical distribution function Thm 1.22, second half — “left for the reader”

Process. $\sqrt{n}(H^{(n)} - H)$ in $\ell^\infty([0, \infty)^2)$ for pairs (Z_i, W_i) .

Rate & class. $r_n = n$, $c_n = 1$, Theorem 1.19.

$$\mathcal{F} = \{f_{s,t}(z, w) = \mathbf{1}_{[0,s]}(z)\mathbf{1}_{[0,t]}(w) : (s, t) \in [0, \infty)^2\}, \quad F \equiv 1.$$

Brackets. Marginal grids $\{s_i\}$, $\{t_j\}$; on the rectangle $[s_{i-1}, s_i] \times [t_{j-1}, t_j]$ take $l_{ij} = \mathbf{1}_{[0, s_{i-1}]} \otimes \mathbf{1}_{[0, t_{j-1}]}$ and $u_{ij} = \mathbf{1}_{[0, s_i]} \otimes \mathbf{1}_{[0, t_j]}$. Monotonicity in each argument gives the sandwich.

L^2 . $u_{ij} - l_{ij}$ is the indicator of an **L-shaped** region, which is contained in the union of the two strips, so

$$\|u_{ij} - l_{ij}\|^2 = H(s_i, t_j) - H(s_{i-1}, t_{j-1}) \leq P(s_{i-1} \leq Z < s_i) + P(t_{j-1} \leq W < t_j).$$

Grid & count. Make every *marginal* strip carry mass $\leq \varepsilon^2/2$; each marginal grid then needs $\leq \lceil 4/\varepsilon^2 \rceil$ points, so

$$N_{[]}(\varepsilon, \mathcal{F}, L^2(P)) \leq pq \leq \lceil 4/\varepsilon^2 \rceil^2 \leq C\varepsilon^{-4}.$$

Finish. $\log N \leq \log C + 4 \log(1/\varepsilon)$, integral $\leq \sqrt{\log C} + \sqrt{\pi} < \infty$; Donsker, covariance $H(s \wedge u, t \wedge v) - H(s, t)H(u, v)$.

Transferable move. Handle dimension by a *product* of grids and bound the L-shape by the union of strips. In d dimensions, $N \leq C\varepsilon^{-2d}$ — still polynomial, still fine.

Example 3 — Censored claims: the sub-distribution process archetype A; a 0/1 passenger

Process. (T_i, δ_i) i.i.d., $H_1(t) = P(T \leq t, \delta = 1)$; $\sqrt{n}(\frac{1}{n} \sum_i \mathbf{1}_{(T_i \leq t)} \delta_i - H_1(t))$ on $[0, \infty)$.

Rate & class. No localisation: $r_n = n$, $c_n = 1$, Theorem 1.19.

$$\mathcal{F} = \{f_t(\tau, d) = \mathbf{1}_{[0, t]}(\tau) d\}$$

Here τ, d are the *slots*; you plug in (T_1, δ_1) . Read the summand off the estimator and replace each random symbol by its slot letter.

Envelope. $\sup_t \mathbf{1}_{[0, t]}(\tau) = 1$ and $d \leq 1$, so $F \equiv 1$, $\mathbb{E}[F^2] = 1$.

Brackets. $u_j - l_j = \mathbf{1}_{[t_{j-1}, t_j]}(\tau) d$. The δ is t -free: it rides along untouched, exactly like the kernel in Example 5.

L^2 . $d \in \{0, 1\} \Rightarrow d^2 = d$, so squaring changes nothing:

$$\|u_j - l_j\|^2 = \mathbb{E}[\mathbf{1}_{(t_{j-1} \leq T_1 < t_j)} \delta_1] = P(t_{j-1} \leq T_1 < t_j, \delta_1 = 1) = H_1(t_j -) - H_1(t_{j-1} -).$$

Grid & count. H_1 is non-decreasing, càdlàg, bounded by 1 — everything Example 1 used about F . So $m \leq \lceil 2/\varepsilon^2 \rceil$, Donsker, covariance $H_1(s \wedge t) - H_1(s)H_1(t)$.

Lesson. *The controlling function need not be the df of the index variable.* Any non-decreasing bounded càdlàg function works — here a sub-distribution function. Same for competing-risks indicators $\mathbf{1}_{(T \leq t)} \mathbf{1}_{(\text{cause}=j)}$.

Example 4 — Inverse-probability-of-censoring weighting archetype A; an unbounded passenger

Setup. $T_i = \min(X_i, C_i)$, $\delta_i = \mathbf{1}_{(X_i \leq C_i)}$, $\bar{G}(y) = P(C_1 > y)$; $\hat{F}_X(t) = \frac{1}{n} \sum_i \mathbf{1}_{(T_i \leq t)} \delta_i / \bar{G}(T_i)$.

Rate & class. $r_n = n$, $c_n = 1$, Theorem 1.19.

$$\mathcal{F} = \left\{ f_t(\tau, d) = \frac{\mathbf{1}_{[0, t]}(\tau) d}{\bar{G}(\tau)} : t \in I \right\}$$

Envelope — and why I is not $[0, \infty)$. With $I = [0, \infty)$, $F(\tau) = d / \bar{G}(\tau) \leq 1 / \bar{G}(\tau)$, and $\bar{G}(\tau) \rightarrow 0$, so $\mathbb{E}[F^2]$ can be infinite. **(1.6) fails.** Cap $I = [0, \tau^*]$: for $\tau > \tau^*$ no $t \in I$ makes the indicator non-zero, so

$$F(\tau) = \frac{\mathbf{1}_{(\tau \leq \tau^*)}}{\bar{G}(\tau)} \leq \frac{1}{\bar{G}(\tau^*)} \quad (\bar{G} \text{ non-increasing}), \quad \mathbb{E}[F^2] \leq \bar{G}(\tau^*)^{-2} < \infty$$

provided $\bar{G}(\tau^*) > 0$. That is the standard hypothesis of every Kaplan–Meier/IPCW theorem, and the envelope is where it comes from.

L^2 . $u_j - l_j = \mathbf{1}_{[t_{j-1}, t_j]}(\tau) d / \bar{G}(\tau)$. Now $1/\bar{G}$ genuinely squares. On the gap, $T_1 < t_j \leq \tau^*$ so $\bar{G}(T_1)^{-2} \leq \bar{G}(\tau^*)^{-2}$:

$$\|u_j - l_j\|^2 \leq \frac{\Delta_j}{\bar{G}(\tau^*)^2}, \quad \Delta_j = H_1(t_j -) - H_1(t_{j-1} -).$$

So $\varphi(\delta) = \sqrt{\delta / \bar{G}(\tau^*)}$, and $\varphi(\delta) \leq \varepsilon$ iff $\Delta_j \leq \varepsilon^2 \bar{G}(\tau^*)^2$.

Count. $m \leq \lceil 2 / (\varepsilon^2 \bar{G}(\tau^*)^2) \rceil = \lceil C / \varepsilon^2 \rceil$. Integral finite; Donsker on $\ell^\infty([0, \tau^*])$ with $\text{Cov} = \mathbb{E}[\mathbf{1}_{(X_1 \leq s \wedge t)} / \bar{G}(X_1)] - F_X(s)F_X(t)$.

Caveat. Real IPCW plugs in an *estimated* $\hat{\bar{G}}$ (Kaplan–Meier), adding a second empirical process and a functional delta-method step (§1.4.3). The bracketing part is exactly the above.

Example 5 — Kernel conditional distribution function Theorem 2.8; archetype A; a changing class

Process. $\sqrt{nh}(F^{(n)}(\cdot | x) - F(\cdot | x))$ in $\ell^\infty([0, \infty))$, $nh^5 \rightarrow 0$, $nh \rightarrow \infty$, $f(x) > 0$.

Reduction. The estimator is a ratio; its denominator is not a process in t . By joint fidi convergence and Theorem 1.8 it suffices to treat the numerator.

Rate & class. $r_n = nh$, $c_n = h^{-1/2}$. Changing class $\Rightarrow$ **Theorem 1.20**.

$$\mathcal{F}_n = \left\{ f_{n,t}(y, z) = h^{-1/2} \mathbf{1}_{[0,t)}(y) K(h^{-1}(x - z)) \right\}$$

Envelope. $h^{-1/2}$ and K have no t ; $\sup_t \mathbf{1}_{[0,t)}(y) = 1$. So $F_n(y, z) = h^{-1/2} K(h^{-1}(x - z))$ — note y drops out entirely. By the kernel-window lemma

$$\mathbb{E}[F_n^2] = \int K^2(u) f(x + hu) du = f(x) \int K^2 + O(h) = O(1). \quad (1.6) \checkmark$$

(1.7). $\mathbb{E}[F_n^3] = h^{-1/2}(f(x) \int K^3 + O(h))$, so the bound is $O(1/(\eta\sqrt{nh})) \rightarrow 0$ since $nh \rightarrow \infty$. $\checkmark$

L^2 . $u_j - l_j = h^{-1/2} \mathbf{1}_{[t_{j-1}, t_j)}(y) K$; squaring and conditioning on X (R3),

$$\|u_j - l_j\|^2 \rightarrow \underbrace{(F(t_j - | x) - F(t_{j-1} | x))}_{\Delta_j} \cdot \underbrace{f(x) \int K^2}_{= \|F_n\|^2},$$

so the localisation constant is $\|F_n\|^2$ and $\varphi(\delta) = \sqrt{\delta}$: $\Delta_j \leq \varepsilon^2$.

Count & finish. $m \leq \lceil C/\varepsilon^2 \rceil$, integral finite (1.9) $\checkmark$. Partition points read off $F(\cdot | x)$, independent of n , so (1.8) is skipped. Limit $(\int K^2/f(x))^{1/2} B \circ F(\cdot | x)$.

Example 6 — Kernel conditional survival function archetype A; the mirror image of Example 5

Process. $\sqrt{nh}(\widehat{S}(\cdot | x) - S(\cdot | x))$ on $[0, \tau]$, numerator $\frac{1}{nh} \sum_i \mathbf{1}_{(Y_i > t)} K(h^{-1}(x - X_i))$.

Rate & class. $r_n = nh$, $c_n = h^{-1/2}$, Theorem 1.20.

$$\mathcal{F}_n = \{f_{n,t}(y, z) = h^{-1/2} \mathbf{1}_{(y > t)} K(h^{-1}(x - z)) : t \in [0, \tau]\}$$

Envelope. $t \mapsto \mathbf{1}_{(y > t)}$ is **non-increasing**, so the sup is at the **left** endpoint $t = 0$; with $Y \geq 0$, $\mathbf{1}_{(y > 0)} = 1$: $F_n(y, z) = h^{-1/2} K(h^{-1}(x - z))$, $\mathbb{E}[F_n^2] \rightarrow f(x) \int K^2$. $\checkmark$

No restriction on I is needed — S is bounded by 1, unlike the tail case.

Brackets. Non-increasing $\Rightarrow$ roles swap: $u_j = f_{n,t_{j-1}}$, $l_j = f_{n,t_j}$, $u_j - l_j = h^{-1/2} \mathbf{1}_{(t_{j-1} < y \leq t_j)} K$.

L^2 , **grid, count.** $\Delta_j = S(t_{j-1} | x) - S(t_j | x)$, $\varphi(\delta) = \sqrt{\delta}$, $\Delta_j \leq \varepsilon^2$, $m \leq \lceil 2/\varepsilon^2 \rceil$. (1.7) iff $nh \rightarrow \infty$; (1.8) skipped.

Example 7 — Tail empirical process Theorem 3.10; archetype A; where I must shrink

Process. $\bar{F} \in RV_{-1/\gamma}$, $u_n \rightarrow \infty$, $k = n\bar{F}(u_n) \rightarrow \infty$; $\tilde{T}_n \Rightarrow W \circ T$ on compacts bounded away from 0, $T(s) = s^{-1/\gamma}$.

Rate & class. $r_n = k$, $c_n = \bar{F}(u_n)^{-1/2}$, Theorem 1.20, $I = [a, b]$ with $a > 0$.

$$\mathcal{F}_n = \{f_{n,t}(z) = \bar{F}(u_n)^{-1/2} \mathbf{1}_{[u_n t, \infty)}(z) : t \in [a, b]\}$$

Check: $c_n n^{-1/2} = (n\bar{F}(u_n))^{-1/2} = k^{-1/2}$, the $\sqrt{k}$ scaling of Def. 3.4.

Envelope, and why $a > 0$. Non-increasing in $t \Rightarrow$ left endpoint: $F_n(z) = \bar{F}(u_n)^{-1/2} \mathbf{1}_{(z > u_n a)}$ and $\mathbb{E}[F_n^2] = \bar{F}(u_n a) / \bar{F}(u_n) \rightarrow a^{-1/\gamma}$. With $a = 0$ this is $1/\bar{F}(u_n) \rightarrow \infty$ and (1.6) fails outright. **“On compacts bounded away from zero” is forced by the envelope**, not a convenience.

(1.7). $\mathbb{E}[F_n^3] / (\eta \sqrt{n}) = a^{-1/\gamma} / (\eta \sqrt{k}) \rightarrow 0$ since $k \rightarrow \infty$.

L^2 . $u_j - l_j = \bar{F}(u_n)^{-1/2} \mathbf{1}_{[u_n t_{j-1}, u_n t_j]}$, so

$$\|u_j - l_j\|^2 = \frac{P(u_n t_{j-1} \leq Z_1 < u_n t_j)}{\bar{F}(u_n)} \rightarrow T(t_{j-1}) - T(t_j) = \Delta_j, \quad \|F_n\|^2 \rightarrow T(a).$$

$\varphi(\delta) = \sqrt{\delta/T(a)}$, so $\Delta_j \leq \varepsilon^2 T(a) = \varepsilon^2 a^{-1/\gamma}$.

Count. Total = $T(a) - T(b) < T(a)$, so $m \leq \lceil C/\varepsilon^2 \rceil - T(a)$ *cancels*. Integral finite. The $o(1)$ is uniform by regular variation, partition n -free, (1.8) skipped.

Example 8 — Conditional tail process Theorem 3.23; archetype A; two localisations at once

Rate & class. $r_n = k_n h_n = n h_n \bar{F}(u_n^x)$, $c_n = (h_n \bar{F}(u_n^x))^{-1/2}$, $I = [a, b]$, $a > 0$.

$$\mathcal{F}_n = \{f_{n,t}(y, z) = (h_n \bar{F}(u_n^x))^{-1/2} \mathbf{1}_{[u_n^x t, \infty)}(y) K(h_n^{-1}(z - x))\}$$

Check: $c_n n^{-1/2} = (k_n h_n)^{-1/2}$. The two normalisations simply multiply.

Envelope. Left endpoint again: $F_n = (h_n \bar{F}(u_n^x))^{-1/2} \mathbf{1}_{(y > u_n^x a)} K(h_n^{-1}(z - x))$, with $\mathbb{E}[F_n^2] \rightarrow g(x) \int K^2 \cdot a^{-1/\gamma(x)} = O(1)$ (Lemma 3.2).

L^2 . Squaring and conditioning on X factorises the two localisations:

$$\|u_j - l_j\|^2 \rightarrow (T^x(t_{j-1}) - T^x(t_j)) \cdot g(x) \int K^2, \quad \|F_n\|^2 \rightarrow T^x(a) \cdot g(x) \int K^2.$$

The whole constant $g(x) \int K^2$ cancels: $\Delta_j \leq \varepsilon^2 T^x(a)$, and $m \leq \lceil C/\varepsilon^2 \rceil$.

Note on constants. The notes carry an extra $1/(2g(x))$ in the stated grid condition; it cancels if you track it through $\|F_n\|^2$, and it changes nothing. Different write-ups of these four proofs carry different constants precisely because constants are free.

Finish. (1.7) iff $k_n h_n \rightarrow \infty$; $o(1)$ uniform by Thm 3.20, so (1.8) skipped; $\Gamma(s, t) = (s \vee t)^{-1/\gamma(x)} g(x) \int K^2$.

Example 9 — Excess-of-loss layer, localised by policy age archetype B inside Theorem 1.20

Setup. (A_i, C_i) i.i.d., age A_i with density f_A , claim $C_i \geq 0$; at fixed age a , $m(t | a) = \mathbb{E}[(C_1 - t)_+ | A_1 = a]$ for $t \in [0, \tau]$; numerator $\frac{1}{nh} \sum_i (C_i - t)_+ K(h^{-1}(a - A_i))$.

Rate & class. $r_n = nh$, $c_n = h^{-1/2}$, Theorem 1.20.

$$\mathcal{F}_n = \{f_{n,t}(c, \alpha) = h^{-1/2} (c - t)_+ K(h^{-1}(a - \alpha)) : t \in [0, \tau]\}$$

Envelope — the twist. $t \mapsto (c - t)_+$ is non-increasing, so the sup is at $t = 0$, and its value is $(c - 0)_+ = c$ — *not* 1. The t -dependent factor no longer disappears:

$$F_n(c, \alpha) = h^{-1/2} c K(h^{-1}(a - \alpha)) \quad \text{— unbounded.}$$

(1.6) and the hypothesis it forces. Condition on the age (R3), then the kernel window:

$$\mathbb{E}[F_n^2] \rightarrow \mathbb{E}[C_1^2 \mid A_1 = a] \cdot f_A(a) \int K^2 \implies \boxed{\mathbb{E}[C_1^2 \mid A_1 = a] < \infty}$$

A finite conditional second moment of the claim size. For Pareto claims with $\alpha \leq 2$ this fails and there is no $\sqrt{nh}$ functional CLT.

Brackets — archetype B. $x \mapsto (x)_+$ is 1-Lipschitz, so for $s < t$, $0 \leq (c-s)_+ - (c-t)_+ \leq t-s$ (it saturates when $c > t$). Hence

$$u_j - l_j \leq (t_j - t_{j-1}) \cdot h^{-1/2} K, \quad \|u_j - l_j\|^2 \leq (t_j - t_{j-1})^2 f_A(a) \int K^2.$$

Grid & count. $\|F_n\|^2 \rightarrow \mathbb{E}[C^2 \mid a] f_A(a) \int K^2$, so $f_A(a) \int K^2$ cancels:

$$\varphi(\delta) = \frac{\delta}{\sqrt{\mathbb{E}[C^2 \mid A = a]}}, \quad t_j - t_{j-1} \leq \varepsilon \sqrt{\mathbb{E}[C^2 \mid A = a]}, \quad m \leq \left\lceil \frac{\tau}{\varepsilon \sqrt{\mathbb{E}[C^2 \mid a]}} \right\rceil = \left\lceil \frac{C}{\varepsilon} \right\rceil.$$

Mesh $\propto \varepsilon$, not ε^2 — and an *equally spaced* grid works, so it is trivially n -free and (1.8) is skipped.

Why this example matters. You were in Theorem 1.20 the whole time and the grid was ε . *Theorem 1.20 does not mean ε^2 .*

Example 10 — Tail moment / mean-excess process archetype A with a weighted controlling function

Setup. $\bar{F} \in RV_{-1/\gamma}$, $\beta := 1/\gamma$, $k = n\bar{F}(u_n) \rightarrow \infty$; $\hat{M}_n(t) = \frac{1}{k} \sum_i \frac{Z_i}{u_n} \mathbf{1}_{(Z_i > u_n t)}$, $t \in [a, b]$, $a > 0$. Karamata: for $\beta > 2$,

$$\frac{1}{\bar{F}(u_n)} \mathbb{E} \left[\left(\frac{Z_1}{u_n} \right)^2 \mathbf{1}_{(Z_1 > u_n t)} \right] \longrightarrow M(t) = \frac{\beta}{\beta - 2} t^{2-\beta},$$

which is finite, positive and *decreasing* on $[a, b]$.

Rate & class. $r_n = k$, $c_n = \bar{F}(u_n)^{-1/2}$, Theorem 1.20.

$$\mathcal{F}_n = \left\{ f_{n,t}(z) = \bar{F}(u_n)^{-1/2} \frac{z}{u_n} \mathbf{1}_{(z > u_n t)} : t \in [a, b] \right\}$$

Envelope. Non-increasing in $t \Rightarrow$ left endpoint: $F_n(z) = \bar{F}(u_n)^{-1/2} (z/u_n) \mathbf{1}_{(z > u_n a)}$. *The passenger z/u_n lives in the same coordinate as the index and is unbounded* — new relative to Examples 5–8.

(1.6) and the hypothesis. $\mathbb{E}[F_n^2] \rightarrow M(a) < \infty$ iff $\beta > 2$, i.e.

$$\boxed{\gamma < 1/2}$$

finite variance in the tail. This is the standard condition for asymptotic normality of tail moment and expected-shortfall estimators; for $\gamma \geq 1/2$ there is no $\sqrt{k}$ CLT.

(1.7) — do not use the third-moment shortcut. It would demand $\beta > 3$. Instead, $F_n > \eta\sqrt{n} \iff Z_1/u_n > \eta\sqrt{n\bar{F}(u_n)} = \eta\sqrt{k}$, so with $V = Z_1/u_n$,

$$\mathbb{E}[F_n^2 \mathbf{1}_{(F_n > \eta\sqrt{n})}] = \frac{\mathbb{E}[V^2 \mathbf{1}_{(V > \eta\sqrt{k})}]}{\bar{F}(u_n)} \approx M(\eta\sqrt{k}) = \frac{\beta}{\beta - 2} (\eta\sqrt{k})^{2-\beta} \rightarrow 0$$

under $\beta > 2$ alone. Note the shape: *the truncation level diverges because $k \rightarrow \infty$.*

L^2 — **still archetype A.** $u_j - l_j = \bar{F}(u_n)^{-1/2}(z/u_n)\mathbf{1}_{(u_n t_{j-1} < z \leq u_n t_j)}$ is supported on a gap, so

$$\|u_j - l_j\|^2 \rightarrow M(t_{j-1}) - M(t_j) = \Delta_j, \quad \|F_n\|^2 \rightarrow M(a), \quad \varphi(\delta) = \sqrt{\delta/M(a)}.$$

The controlling function is no longer a distribution function; it is a weighted tail moment M . All the construction ever needed was that M be monotone and finite on I .

Count. $\text{Total} = M(a) - M(b) < M(a)$, so $m \leq \lceil 2/\varepsilon^2 \rceil - M(a)$ cancels. (1.8) skipped, grid read off M .

Example 11 — Exponential tilts archetype B; Theorem 1.19; a smooth one-parameter class

Claim. $Z_i \geq 0$ i.i.d. with $\mathbb{E}[Z_1^2] < \infty$. Then $\mathcal{F} = \{f_\theta(z) = e^{-\theta z} : \theta \in [0, 1]\}$ is P -Donsker.

Envelope. $\theta \mapsto e^{-\theta z}$ is non-increasing on $z \geq 0$, sup at $\theta = 0$: $F \equiv 1$.

Lipschitz majorant (R4). By the mean value theorem, for $z \geq 0$,

$$|e^{-\theta z} - e^{-\theta' z}| \leq |\theta - \theta'| z e^{-(\theta \wedge \theta') z} \leq |\theta - \theta'| z =: |\theta - \theta'| m(z),$$

and $m \in L^2(P)$ exactly because $\mathbb{E}[Z_1^2] < \infty$.

Brackets. Grid $0 = \theta_0 < \dots < \theta_m = 1$ with mesh $\leq \varepsilon/\|Z_1\|_{L^2(P)}$; monotonicity gives $l_j = f_{\theta_{j+1}}$, $u_j = f_{\theta_j}$, and $u_j - l_j \leq (\theta_{j+1} - \theta_j)m$ pointwise, so $\|u_j - l_j\|_{L^2(P)} \leq (\theta_{j+1} - \theta_j)\|Z_1\|_{L^2(P)} \leq \varepsilon$.

Count. $N_{[]}(\varepsilon, \mathcal{F}, L^2(P)) \leq \lceil \|Z_1\|_{L^2(P)}/\varepsilon \rceil$; $\log N \leq \log(1 + \|Z_1\|_{L^2(P)}/\varepsilon)$, integral finite, Donsker; and Glivenko–Cantelli free via $L^1 \leq L^2$.

The point. The class is monotone — so you get the sandwich for free — but the bracket size is governed by R4, not R2, because $u - l$ is not an indicator. Grid ε , not ε^2 . *Monotone $\neq$ archetype A.*

Example 12 — Linear and affine classes Examples 1.8–1.9 in the notes; archetype B; where ε^{-2} means something else

(a) One parameter. $\mathcal{F}_1 = \{f_\theta(z) = \theta z : \theta \in [-1, 1]\}$, $\mathbb{E}[Z_1^2] < \infty$. Grid of mesh $\leq \varepsilon/\|Z_1\|_{L^2(P)}$; on $[\theta_j, \theta_{j+1}]$ take $l_j = \min\{f_{\theta_j}, f_{\theta_{j+1}}\}$, $u_j = \max\{\cdot\}$ (R5). Then $\|u_j - l_j\|_{L^2(P)} = (\theta_{j+1} - \theta_j)\|Z_1\|_{L^2(P)} \leq \varepsilon$ and

$$N_{[]}(\varepsilon, \mathcal{F}_1, L^2(P)) \leq \lceil 2\|Z_1\|_{L^2(P)}/\varepsilon \rceil.$$

(b) Two parameters. $\mathcal{F}_2 = \{a + bz : (a, b) \in [-1, 1]^2\}$. Rectangles with $\Delta_a = \varepsilon/2$, $\Delta_b = \varepsilon/(2\|Z_1\|_{L^2(P)})$; extrema at the corners give $u - l \leq \Delta_a + \Delta_b|z|$, so $\|u - l\|_{L^2(P)} \leq \Delta_a + \Delta_b\|Z_1\|_{L^2(P)} \leq \varepsilon$, and the number of rectangles is

$$N_{[]}(\varepsilon, \mathcal{F}_2, L^2(P)) \leq \lceil 2/\Delta_a \rceil \lceil 2/\Delta_b \rceil \leq C\varepsilon^{-2}.$$

Read the exponent correctly. This ε^{-2} comes from having **two parameters** ($(C/\varepsilon)^d$ with $d = 2$), *not* from a square root. Archetype A with one index gets ε^{-2} from R2. Same exponent, different mechanism — being able to say which is the test of whether you have understood §6.

Example 13 — Stop-loss above a deductible archetype A; unbounded passenger in the index coordinate

Setup. $C_1, C_2, \dots \geq 0$ i.i.d. claim severities, F_C continuous. The expected loss from claims piercing a deductible t is $L(t) = \mathbb{E}[C_1 \mathbf{1}_{(C_1 > t)}]$, estimated by $\hat{L}_n(t) = \frac{1}{n} \sum_i C_i \mathbf{1}_{(C_i > t)}$. Show $\sqrt{n}(\hat{L}_n - L) \Rightarrow \mathbf{G}$ in $\ell^\infty([0, \infty))$.

Rate & class. Every observation enters the average: $r_n = n$, $c_n = 1$. Fixed class, $\sqrt{n}$ rate $\Rightarrow$ **Theorem 1.19**.

$$\mathcal{F} = \{f_t(c) = c \mathbf{1}_{(c>t)} : t \in [0, \infty)\}, \quad w(c) = c, \quad g_t(c) = \mathbf{1}_{(c>t)}.$$

Envelope, and the assumption it forces. The carrier is non-increasing in t (raise the deductible, fewer claims pierce it), so the sup is at the *left* endpoint $t = 0$, where $\mathbf{1}_{(c>0)} = 1$ — but the passenger stays:

$$F(c) = c, \quad \mathbb{E}[F^2] = \mathbb{E}[C_1^2] < \infty \text{ must be imposed.}$$

The problem gave only continuity of F_C , so this is a new hypothesis.

Shrinking I cannot fix a bad data tail

Contrast Example 7, where $\mathbb{E}[F_n^2] \rightarrow \infty$ as $a \downarrow 0$ and cutting I away from the boundary repaired it. Here the divergence comes from the *data's* tail, not from the index: restricting to $t \geq a$ gives envelope $c \mathbf{1}_{(c>a)}$ and $\mathbb{E}[C^2 \mathbf{1}_{(C>a)}]$, still infinite if $\mathbb{E}[C^2]$ is. **Index-set surgery helps only when the index causes the divergence**; otherwise you impose a moment condition.

Brackets. Non-increasing $\Rightarrow$ largest member at the left end of the cell:

$$u_j(c) = c \mathbf{1}_{(c>t_{j-1})}, \quad l_j(c) = c \mathbf{1}_{(c>t_j)}, \quad u_j - l_j = c \mathbf{1}_{(t_{j-1}<c\leq t_j)}.$$

L^2 — and a trap. Squaring gives $c^2 \mathbf{1}_{\text{gap}}$, so

$$\|u_j - l_j\|^2 = \mathbb{E}[C_1^2 \mathbf{1}_{(t_{j-1}<C_1\leq t_j)}] = \int_{t_{j-1}}^{t_j} c^2 dF_C(c).$$

This is *not* $\mathbb{E}[C_1^2](F_C(t_j) - F_C(t_{j-1}))$: the passenger and the carrier are functions of the **same** variable, so they cannot be separated. Compare Example 5, where K lived in X and the indicator in Y , and conditioning genuinely factorised them.

Controlling function. With $\Lambda(t) = \|f_t\|^2 = \mathbb{E}[C_1^2 \mathbf{1}_{(C_1>t)}] = \int_t^\infty c^2 dF_C$,

$$\int_{t_{j-1}}^{t_j} = \int_{t_{j-1}}^\infty - \int_{t_j}^\infty \quad \Rightarrow \quad \|u_j - l_j\|^2 = \Lambda(t_{j-1}) - \Lambda(t_j) = \Delta_j.$$

Λ is *not* a probability — it is the second moment restricted to claims above t . It is non-increasing, $\Lambda(0) = \mathbb{E}[C_1^2]$, $\Lambda(\infty) = 0$.

Grid & count. Theorem 1.19, so the tolerance is a plain ε : $\Delta_j \leq \varepsilon^2$. The increments telescope, $\sum_{j=1}^m \Delta_j = \Lambda(0) - \Lambda(\infty) = \mathbb{E}[C_1^2]$, so with each step of size ε^2 ,

$$m \varepsilon^2 \approx \mathbb{E}[C_1^2] \quad \Rightarrow \quad N_{[]}(\varepsilon, \mathcal{F}, L^2(P)) \leq \left\lceil \frac{2 \mathbb{E}[C_1^2]}{\varepsilon^2} \right\rceil.$$

You never evaluate Δ_j — you choose the cut points so that it is small. Picture it as slicing the **range** of Λ (a drop of total height $\mathbb{E}[C_1^2]$) into steps of height ε^2 , not slicing the domain.

Finish. Integral finite, P -Donsker, $\text{Cov}(\mathbf{G}(s), \mathbf{G}(t)) = \Lambda(s \vee t) - L(s)L(t)$; and $L^1 \leq L^2$ gives Glivenko-Cantelli free.

Example 14 — Conditional stop-loss at a fixed policy age the same, localised: two passengers, one factorises and one does not

Setup. (A_i, C_i) i.i.d., age density f_A with $f_A(a) > 0$ and continuous at a ; $h = h_n \rightarrow 0$, $nh \rightarrow \infty$. With $L(t | a) = \mathbb{E}[C_1 \mathbf{1}_{(C_1 > t)} | A_1 = a]$ and numerator $\frac{1}{nh} \sum_i C_i \mathbf{1}_{(C_i > t)} K(h^{-1}(a - A_i))$, prove weak convergence of $\sqrt{nh}(\hat{L}_n(\cdot | a) - \mathbb{E}\hat{L}_n(\cdot | a))$ in $\ell^\infty([0, \infty))$.

Rate & class. $r_n = nh$, $c_n = h^{-1/2}$. A bandwidth appears $\Rightarrow$ **Theorem 1.20**.

$$\mathcal{F}_n = \{f_{n,t}(c, \alpha) = h^{-1/2} c \mathbf{1}_{(c > t)} K(h^{-1}(a - \alpha)) : t \in [0, \infty)\}$$

Two passengers, c and K ; one carrier, $\mathbf{1}_{(c > t)}$.

Envelope. Both passengers ride outside the sup; carrier maximal at $t = 0$:

$$F_n(c, \alpha) = h^{-1/2} c K(h^{-1}(a - \alpha)).$$

Conditioning on A_1 (legitimate: C and K are in different coordinates) then applying the kernel window,

$$\mathbb{E}[F_n^2] = \mathbb{E}[\mathbb{E}[C_1^2 | A_1] h^{-1} K^2] \rightarrow \mathbb{E}[C_1^2 | A_1 = a] f_A(a) \int K^2.$$

Of the three factors, $f_A(a)$ and $\int K^2$ were given; so the hypothesis to impose is $\mathbb{E}[C_1^2 | A_1 = a] < \infty$. (1.6) ✓

(1.7). $\mathbb{E}[F_n^3] = h^{-1/2}(\mathbb{E}[C_1^3 | A_1 = a] f_A(a) \int K^3 + O(h))$, so the bound is $O(1/(\eta \sqrt{nh})) \rightarrow 0$ since $nh \rightarrow \infty$. Note this shortcut costs *one more* assumption than (1.6) did: $\mathbb{E}[C_1^3 | A_1 = a] < \infty$.

L^2 — **which passenger factorises?** $u_j - l_j = h^{-1/2} c \mathbf{1}_{(t_{j-1} < c \leq t_j)} K$, and after squaring and conditioning on A_1 ,

$$\|u_j - l_j\|^2 \longrightarrow \underbrace{\left(L_2(t_{j-1} | a) - L_2(t_j | a) \right)}_{\text{C stays inside: same coordinate as the carrier}} \cdot \underbrace{f_A(a) \int K^2}_{\text{K factorises out: different coordinate}},$$

where $L_2(t | a) := \mathbb{E}[C_1^2 \mathbf{1}_{(C_1 > t)} | A_1 = a]$. So $\Lambda_n(t) = \|f_{n,t}\|^2 \rightarrow L_2(t | a) f_A(a) \int K^2$ and $\|F_n\|^2 \rightarrow L_2(0 | a) f_A(a) \int K^2$ with $L_2(0 | a) = \mathbb{E}[C_1^2 | A_1 = a]$.

Grid & count. The whole constant $f_A(a) \int K^2$ cancels in the ratio:

$$\frac{\|u_j - l_j\|}{\|F_n\|} = \sqrt{\frac{\Delta_j^{L_2}}{L_2(0 | a)}} \leq \varepsilon \iff \Delta_j^{L_2} \leq \varepsilon^2 L_2(0 | a), \quad m \leq \left\lceil \frac{2L_2(0 | a)}{\varepsilon^2 L_2(0 | a)} \right\rceil = \left\lceil \frac{2}{\varepsilon^2} \right\rceil.$$

Integral finite (1.9) ✓; grid read off $L_2(\cdot | a)$, which does not move with n , so (1.8) skipped. Limit covariance $\Gamma(s, t) = L_2(s \vee t | a) f_A(a) \int K^2$.

Why the constant survived in Example 13 but cancelled in Example 14

It is not the kernel. It is the **tolerance**.

	Example 13 (Thm 1.19)	Example 14 (Thm 1.20)
tolerance	ε — <i>absolute</i>	$\varepsilon \ F_n\ _{L^2(p)}$ — <i>relative</i>
grid condition	$\Delta_j \leq \varepsilon^2$	$\Delta_j \leq \varepsilon^2 \Lambda(0)$
count	$\Lambda(0)/\varepsilon^2$ — survives	$\Lambda(0)/(\varepsilon^2 \Lambda(0))$ — cancels

With an *absolute* tolerance the per-cell budget is a fixed ε^2 , so the total budget $\Lambda(0)$ divides through and appears in the count. With a *relative* tolerance the per-cell budget is itself proportional to $\Lambda(0)$, which therefore appears top and bottom and vanishes.

And that is precisely why Theorem 1.20 is stated with $\varepsilon \|F_n\|$. For a changing class $\Lambda(0)$ contains h^{-1} , $\bar{F}(u_n)^{-1}$ and friends — quantities that blow up with n . An absolute tolerance would give a bracket count growing in n , and $\int_{[0, \delta_n]} \sqrt{\log N_{[]} } d\varepsilon$ could never tend to zero. For a fixed class there is nothing to protect against, so the plain ε is fine and the surviving constant is harmless.

Part III — Reference

9 Master table

Example	Thm	Class $f_{n,t}$	Envelope F_n	$\lim \Lambda_n$	N
1 empirical df	1.19	$\mathbf{1}_{[0,t]}(z)$	1	F	$C\varepsilon^{-2}$
2 bivariate df	1.19	$\mathbf{1}_{[0,s]}\mathbf{1}_{[0,t]}$	1	F_Z, F_W	$C\varepsilon^{-4}$
3 censored H_1	1.19	$\mathbf{1}_{[0,t]}(\tau)d$	1	H_1	$C\varepsilon^{-2}$
4 IPCW	1.19	$\mathbf{1}_{[0,t]}(\tau)d/\overline{G}(\tau)$	$1/\overline{G}(\tau^*)$	H_1	$C\varepsilon^{-2}$
5 kernel cdf	1.20	$h^{-1/2}\mathbf{1}_{[0,t]}(y)K$	$h^{-1/2}K$	$F(\cdot x)$	$C\varepsilon^{-2}$
6 kernel surv.	1.20	$h^{-1/2}\mathbf{1}_{(y>t)}K$	$h^{-1/2}K$	$S(\cdot x)$	$C\varepsilon^{-2}$
7 tail process	1.20	$\overline{F}(u_n)^{-1/2}\mathbf{1}_{(z>u_nt)}$	$\overline{F}^{-1/2}\mathbf{1}_{(z>u_na)}$	T	$C\varepsilon^{-2}$
8 cond. tail	1.20	$(h\overline{F})^{-1/2}\mathbf{1}_{(y>u_n^*t)}K$	at $t = a$	T^x	$C\varepsilon^{-2}$
9 XL layer	1.20	$h^{-1/2}(c - t)_+K$	$h^{-1/2}cK$	mesh	$C\varepsilon^{-1}$
10 tail moment	1.20	$\overline{F}^{-1/2}\frac{z}{u_n}\mathbf{1}_{(z>u_nt)}$	at $t = a$	M	$C\varepsilon^{-2}$
11 exp. tilts	1.19	$e^{-\theta z}$	1	mesh	$C\varepsilon^{-1}$
12 affine	1.19	$a + bz$	$1 + z $	mesh	$C\varepsilon^{-2}$
13 stop-loss	1.19	$c\mathbf{1}_{(c>t)}$	c	$\mathbb{E}[C^2\mathbf{1}_{(C>t)}]$	$C\varepsilon^{-2}$
14 cond. stop-loss	1.20	$h^{-1/2}c\mathbf{1}_{(c>t)}K$	$h^{-1/2}cK$	$L_2(t a)f_A(a)\int K^2\varepsilon^{-2}$	

Read down the $\lim \Lambda_n$ column: $F, F_Z, H_1, F(\cdot|x), S(\cdot|x), T, T^x, M$. Distribution functions, sub-distribution functions, tail functions, weighted moments — *the only property used is monotone and finite on I* . And you never have to recognise which of them applies: by §3, $\Lambda_n(t) = \|f_{n,t}\|_{L^2(P)}^2$, so you get it by computing the squared norm of one class member as a function of t .

10 Write-up template — Theorem 1.20

Six blanks; everything else is verbatim

Setup. The limit covariance is identified by a finite-dimensional argument (Lindeberg + Cramér–Wold), giving $\Gamma(s, t) = [1]$. To prove functional convergence we apply the Bracketing CLT for changing functions, Theorem 1.20. Consider $I = [2]$ and define $\mathcal{F}_n = \{f_{n,t} = c_n g_t : t \in I\}$ with $c_n = [3]$.

Envelope. By monotonicity in t , an envelope is $F_n = [4]$. By [kernel expansion / regular variation], $\mathbb{E}[F_n^2] = [5] = O(1)$, showing (1.6).

Lindeberg. Since $\mathbf{1}_{(F_n > \eta\sqrt{n})} \leq F_n / (\eta\sqrt{n})$, $\mathbb{E}[F_n^2 \mathbf{1}_{(F_n > \eta\sqrt{n})}] \leq \mathbb{E}[F_n^3] / (\eta\sqrt{n}) = O(1/\sqrt{r_n}) \rightarrow 0$, since $r_n = [6] \rightarrow \infty$. This shows (1.7).

Brackets. (Recall $\Delta_j = |\Lambda_n(t_j) - \Lambda_n(t_{j-1})|$ and $\Delta_{\text{env}} = \Lambda_n(t^*)$ with $\Lambda_n(t) = \|f_{n,t}\|_{L^2(P)}^2$.) For $j = 1, \dots, m$ and points $t_0 < \dots < t_m$ spanning I , let l_j, u_j be $f_{n,t}$ at the two endpoints of cell j , and $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$, so that $\mathcal{F}_n = \bigcup_j \mathcal{F}_j^n$ and $\sup_{f, g \in \mathcal{F}_j^n} |f - g| \leq u_j - l_j$ pointwise. Let $\varepsilon > 0$ and choose $m \leq \lceil 2/\varepsilon^2 \rceil$ points with $\Delta_j \leq \frac{1}{2}\varepsilon^2 \Delta_{\text{env}}$, possible because [the controlling function is càdlàg, monotone and bounded on I]. Then $\|u_j - l_j\|_{L^2(P)} = \sqrt{\Delta_j} (\text{const})^{1/2} (1 + o(1)) \leq \varepsilon \|F_n\|_{L^2(P)}$, so $N_{[]}(\varepsilon \|F_n\|_{L^2(P)}, \mathcal{F}_n, L^2(P)) \leq m \leq \lceil C/\varepsilon^2 \rceil$.

Entropy. Hence $\int_0^1 \sqrt{\log N_{[]} d\varepsilon} \leq \int_0^1 \sqrt{\log(C/\varepsilon^2)} d\varepsilon < \infty$, and by absolute continuity of the integral $\int_{[0, \delta_n]} \rightarrow 0$ for every $\delta_n \rightarrow 0$, showing (1.9).

Closing. The $o(1)$ above is uniform in the partition, so the partition points may be chosen independently of n and (1.8) is unnecessary. Since the covariances converge pointwise to Γ , Theorem 1.20 yields $\mathbf{G}^{(n)} f_{n,t} \Rightarrow$ a tight Gaussian process in $\ell^\infty(I)$ with a continuous version. $\square$

11 Write-up template — Theorem 1.19

Shorter. Define $\mathcal{F}$. Build the brackets as above — no c_n , no envelope scaling, so the tolerance is a plain ε and the grid condition is $\Delta_j \leq \varepsilon^2$ (archetype A) or $\text{mesh} \leq \varepsilon / \|m\|_{L^2(P)}$ (archetype B). Bound $N_{[]}(\varepsilon, \mathcal{F}, L^2(P)) \leq C\varepsilon^{-p}$. Show $J_{[]}(\mathbf{1}, \mathcal{F}, L^2(P)) < \infty$. Conclude P -Donsker. Then add the free line: “since $\|g\|_{L^1} \leq \|g\|_{L^2}$, $N_{[]}(\varepsilon, \mathcal{F}, L^1(P)) < \infty$ for every ε , so $\mathcal{F}$ is also P -Glivenko–Cantelli by Theorem 1.18.”

12 Traps, ranked by how often they cost marks

1. **Wrong normalisation.** Symptom: $\mathbb{E}[F_n^2]$ is 0 or ∞ instead of a constant. Audit (1.6) immediately after writing the class.
2. **Dropping the $\|F_n\|$.** Theorem 1.20 uses $\varepsilon \|F_n\|_{L^2(P)}$. Writing a bare ε stops the constants cancelling and the algebra explodes.
3. **Index set too large.** Tail-type processes need I bounded away from the boundary where the envelope blows up. State *why*; don’t just copy $[a, b]$.
4. **Confusing ε with ε^2 .** ε^2 is a *mass*; ε is a *distance*. Decide by looking at what $u_j - l_j$ is.
5. **Verifying (1.8) unnecessarily.** Five minutes you don’t have. One sentence instead.

6. **Claiming brackets belong to $\mathcal{F}$.** They need not, and usually don't ($\mathbf{1}_{[0,t_{j-1}]}$ is left-open while the class has closed indicators). That is the difference between *bracketing* and *covering* numbers — don't "fix" it.
7. **Forgetting the fidis.** Bracketing gives tightness; it does not identify the limit.
8. **Brackets that still contain t .** l_j, u_j must be two *fixed* functions sandwiching the whole cell.
9. **Ignoring atoms.** Use $F(t_j-)$ and the greedy construction; if F is continuous say so explicitly and simplify.
10. **Optimising constants.** Only the power of ε reaches the conclusion.

13 Exam plan

Memorise verbatim (about fifteen minutes' work)

1. Theorem 1.20: four conditions, the covariance proviso, and the final sentence about (1.8).
2. $c_n = \sqrt{n/r_n}$.
3. $\mathbf{1}_{(F_n > \eta\sqrt{n})} \leq F_n / (\eta\sqrt{n})$.
4. $\|\mathbf{1}_B - \mathbf{1}_A\|_{L^2(P)}^2 = P(B \setminus A)$ for $A \subseteq B$.
5. $\mathbb{E}[h^{-1}K^p(h^{-1}(x - X))] = f(x) \int K^p + O(h)$.
6. $\int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon = \Gamma(3/2) < \infty$, and the phrase "by absolute continuity of the integral".
7. The three closing sentences: partition n -free $\Rightarrow$ skip (1.8); $L^1 \leq L^2 \Rightarrow$ Glivenko–Cantelli free; tight Gaussian limit has a continuous version.

Derive, never memorise

The envelope (it's a sup — take it), $\mathbb{E}[F_n^2]$ (one of two lemmas), the bracket pair (cell endpoints), the grid condition (square the target), the count (divide the total budget).

A twenty-minute answer, budgeted

Min	Do	Marks it protects
0–2	strip t -free parts; name r_n, c_n ; write $\mathcal{F}_n$ and I	setup — usually a third of the marks and the part everyone rushes
2–4	envelope by monotonicity; $\mathbb{E}[F_n^2] = O(1)$ (1.6)	also validates c_n ; errors caught here are cheap
4–6	Markov / third moment (1.7)	free marks, one displayed line
6–13	brackets, L^2 , grid condition, count	the core — show the $\sqrt{\Delta_j}$ line explicitly, that is where the marker looks
13–16	entropy integral, then the δ_n remark (1.9)	two lines, easy to forget under pressure
16–18	the (1.8) sentence; state the covariance from the fidis	completeness marks
18–20	conclude in $\ell^\infty(I)$, name the limit, add the free G–C line	the conclusion is a mark in itself

Three questions to ask of any new scenario, in order

1. **What is the effective sample size r_n ?** It is n times the probability of the localising event. It gives c_n , it tells you what (1.7) reduces to, and it is usually the rate you were asked to prove.
2. **Is the t -dependence an indicator or a smooth function?** Indicator $\Rightarrow$ archetype A, mass ε^2 . Smooth $\Rightarrow$ archetype B, mesh ε .
3. **What monotone bounded function controls the mass?** $F, F(\cdot \mid x), T, T^x, H_1, S(\cdot \mid x), M, \dots$ Its total variation on I must be finite — that, and only that, is what makes the grid exist.

If you get stuck mid-proof

Almost every dead end is one of three things: the envelope's second moment diverges (*shrink I*); the constants won't cancel (*you dropped a factor from $\|F_n\|$*); or you can't make the gaps small (*your controlling function has infinite variation on I — shrink I*). Check those before rethinking the strategy. The strategy is almost never the problem.

Sources. M. Bladt, *Statistics for Insurance: Theory and methods for life and non-life insurance data*, version of 14 September 2026: Definitions 1.14–1.17, Theorems 1.18–1.22, Examples 1.8–1.9, Theorem 2.8, Theorems 3.10 and 3.23, Lemma 3.3. Examples 2, 3, 4, 6, 9, 10 and 11 are constructed in the notes' own style; Example 2 is the second half of Theorem 1.22, which the text leaves to the reader.